

[9]

SARDAR PATEL UNIVERSITY

M.SC(IT) Sem-II EXAMINATION

2016

THURSDAY, 31th MARCH

10:30 am to 12:30 pm

SUBJECT: MATHEMATICS-II (PS02FIIT02)

Maximum Marks: 70

Q-1 Write the correct option in the answer book.

[10]

- (1) The set $\{x \in \mathbb{Z} : -500 < x < 500\}$ is:
 - (a) finite
 - (b) Infinite
 - (c) Empty
 - (d) none
- (2) The set $\{N, +\}$ is:
 - (a) group
 - (b) ring
 - (c) monoid
 - (d) semi group
- (3) Quartile deviation is defined as
 - (a) $\frac{Q_3 - Q_1}{2}$
 - (b) $\frac{Q_3 - Q_1}{3}$
 - (c) $\frac{Q_1 - Q_2}{2}$
 - (d) $\frac{Q_3 + Q_1}{2}$
- (4) We can select 4 objects from the given 10 objects in..... ways.
 - (a) $\binom{4}{10}$
 - (b) $\binom{10}{4}$
 - (c) $\frac{10!}{4!}$
 - (d) $\frac{9!}{4!}$
- (5) $P(8, 4) =$
 - (a) 1480
 - (b) 1680
 - (c) 168
 - (d) 1860
- (6) Let $A = \{0, 1\}$, then A closed under:
 - (a) multiplication
 - (b) addition
 - (c) Division
 - (d) Subtraction
- (7) Every monoid are:
 - (a) group
 - (b) ring
 - (c) semi group
 - (d) none
- (8) A square of is called Variance
 - (a) Range
 - (b) Mean deviation
 - (c) Standard deviation
 - (d) Quartile deviation
- (9) De Morgan's Law:
 - (a) $(A \cup B)^c = A^c \cap B^c$
 - (b) $(A \cap B)^c = (A \cap B)^c$
 - (c) $(A \cup B)^c = A \cap B$
 - (d) None
- (10) The number of elements in a set $\{x \in \mathbb{Z} : x^2 - 2 = 0\}$ are:
 - (a) 1
 - (b) 2
 - (c) $\pm\sqrt{2}$
 - (d) 0

Q:2 Answer the following in short. (Any Ten)

[20]

- (1) Find variance of the observations: 5, 5, 5, 5, 5, 5, 5, 5.
- (2) Find the power set of a set $\{1, 2, 3\}$.

(PTO)

- (3) Explain the positive correlation with two examples.
- (4) How many committees of six with a given chair person can be selected from twelve persons?
- (5) Find the number of distinct permutation that can be formed from all the letters of the word (1) MATHEMATICS (2) RESERVATION.
- (6) Define the terms: Group, Monoid.
- (7) If $A = \{1, 2, 3, \dots, 12\}$ and $B = \{4, 6, 7, \dots, 16\}$ then find the symmetrical difference between A and B.
- (8) Find dual of the following:
 (i) $(A \cap B \cup C)^c = (A \cap C)^c \cap (A \cap B)^c$, (ii) $(A \cap \phi) \cup (U \cap A^c) = A$.
- (9) Find the number of ways that a party of 8 persons can arrange themselves: (i) around a circular table and (ii) in a row.
- (10) If S is a nonempty set with the operation $a*b = b$. Is the operation* associative?
- (11) Define range and variance.
- (12) In Z_{10} , find $-3, -8, 3^{-1}, 5^{-1}$.

Q-3

- (a) Define a function F as $F(a, b) = 0$ if $a < b$.
 $= F(a - b, b) + 1$ if $b \leq a$.

[05]

Find $F(4, 3)$ and $F(14, 3)$.

- (b) Find inverse of the function $f(x) = \frac{7x-3}{5x-2}$, $x \neq \frac{2}{5}$.

[05]

Q-3

OR

- (c) Prove that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

[05]

- (d) If $f(x) = x^2 + 5x$ and $g(x) = 3x + 2$ then find: (i) fog (ii) fog(-1) (iii) gof(1).

[05]

Q-4

- (a) Show that the set $\{1, 2, 3, 4, 5, 6\}$ is a group under multiplication modulo 7.

[05]

For $a, b \in Q$ (rational numbers), define $a*b = a + b - ab$.

[05]

- (b)
 (i) Is $(Q, *)$ Semigroup?
 (ii) Is $(Q, *)$ Monoid?
 (iii) Find its inverse if it exist.

Q-4

OR

- (c) For a, b rational number set Q , define $a*b = ab/5$. Show that $(Q, *)$ is a group under binary operation *. Is it commutative?

[05]

- (d) For a, b rational number, define $a*b = ab/3$. Is $(Q, *)$ commutative? Show that $(Q, *)$ is Monoid and find its inverse if it exist. [05]

Q-5

- (a) A class consist of seven men and five women. (i) Find the number m of committees of five that can be selected from the class (ii) Find the number m of committees of five if it consists of three men and two women. [05]

- (b) A bag contains five red marbles and six white marbles. Find the number m of ways that four marbles can be drawn from the bag such that: (i) two of the marbles must be red and two of the marbles must be white, (ii) at least one marble is red. [05]

Q-5

OR

- (c) A debating team consists of 3 boys and 3 girls. Find the number of ways they can sit in a row where: (a) there are no restrictions; (b) the boys and girls are each to sit together; (c) just the girls are to sit together. [05]

- (d) Find the number of distinct permutation that can be formed from all the letters of the word MISSISSIPPI. Also find the number of distinct permutation if the words are to begin with I. [05]

- Q.6 Given below is the frequency distribution of the marks obtained by 90 students. [06]

- (a) Compute the standard deviation.

Marks:	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No. of students	5	12	15	20	18	10	6	4

- (b) Given $\sum X = 125$, $\sum Y = 100$, $\sum X^2 = 650$, $\sum Y^2 = 436$, $\sum XY = 520$ and $n = 25$ Obtain the value of Karl Pearson's coefficient of correlation r . [04]

OR

- Q.6 Calculate the correlation coefficient for the following heights (in inches) of [06]

- (a) fathers (X) and their sons (Y):

X:	65	66	67	67	68	69	70	72
Y:	67	68	65	68	72	72	69	71

- (b) Find quartile deviation for the following data [04]

Class	0-15	15-30	30-45	45-60	60-75	75-90	90-105
f	8	26	30	45	20	17	4

$$X = X = X$$